

Tharun Ganeshram

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Technical Skills

Languages: C, C++, Python, Bash, Expect-Lite (TCL extension)

Technologies & Frameworks: Intel TBB, MLIR, CUDA, PyTorch, TensorFlow, OpenCV, Keras, Scikit-learn, pandas

OS & Tools: Linux, FreeRTOS, IAR, MATLAB, Git, Jira, Confluence, SolidWorks, PlantUML, Jenkins, InfluxDB

Education

University of Waterloo

Candidate for Bachelor of Applied Science (B.A.Sc.) in **Honours Systems Design Engineering**

Sep. 2022 – Apr. 2027

GPA: 4.0

Experience

Machine Learning Software Engineer

Jan. 2025 – Present

Cerebras Systems Inc.

Sunnyvale, CA

- Redesigned inference pipeline idle logic using placeholder tokens to **reduce idle sleep & startup delays by 100%**
- **Exponentially increased LLM speculative decoding metrics** via **C++** per-token state tracking & **InfluxDB** output
- Designed prefix search in **Python** for off-wafer prompt caching that **increased inference speed by over 40%**
- Pioneered **multimodal LLM inference** runtime compatibility while maintaining **existing time & space** complexity
- Enabled **16+** on-wafer boolean operations by implementing custom **MLIR** compiler passes, operations, & lowerings

Firmware Team Lead

Dec. 2022 – Present

University of Waterloo Hyperloop Design Team

Waterloo, ON

- **Achieved sub-100ms latency** after system re-architecture using **FreeRTOS** thread & resource management
- Fulfilled battery safety compliance by configuring thermistors and fans to **STM32** & **Raspberry Pi** using **SPI** & **CAN**

Embedded Flight Software Developer

May 2024 – Dec. 2024

Waterloo Aerial Robotics Group

Waterloo, ON

- Architected **37% signal noise reduction** by investigating **STM32 DroneCAN** integration, decoding, & debugging
- Established the conversion of **DroneCAN** servo signals from a **Pixhawk** to PWM motor signals using **STM32** in **C**

Firmware Engineer

May 2024 – Aug. 2024

Blackline Safety Corporation

Waterloo, ON

- Implemented **UART** communication, **BSP** control signal abstraction & sensor classes in **C++** for **5** new EXO8 sensors
- Configured execute-in-place flash memory & clock with screen optimizations for **6x increase** in XIP processing rate

Embedded Software Developer

Jan. 2023 – Apr. 2023

Ciena Corporation

Ottawa, ON

- Engineered **Expect-Lite** to **Python** converter, reducing manual file upgrade time from **3 weeks to 5 hours**
- Modernized legacy **Bash** Sanity systems using **Python**, redesigning logic with modules to **reduce runtime by 80%**

Data Science & Machine Learning Engineer

Dec. 2022 – Apr. 2023

OpenHaus

Ottawa, ON

- Developed data preprocessing pipeline & employed various **Python** machine learning algorithms through **pandas**, **scikit-learn**, **TensorFlow**, and **Keras** to model customer behaviour and predict rental prices with **93% accuracy**

Projects

Omni-Directional Bluetooth Car 🚗 | *Embedded C, STM32, ESP32, UART, SolidWorks*

- Designed car body & firmware using **servo motors**, **STM32**, and **ESP32** for real-time **UART Bluetooth** control

5-Bar Pick and Place Mechanism 🤖 | *Arduino Programming Language, BotBoarduino*

- Established precise payload transportation using **hobby servos**, **BotBoarduinos** & inter-board communication

Real-Time Object Recognition App 📱 | *Python, TensorFlow, OpenCV*

- Trained unsupervised ML model using **TensorFlow** & **OpenCV** to accurately identify & classify objects in live video